Diffie Hellman Key Exch

assume prime no. (q)

Select α → such that — α - primitive root of q

$\alpha < q$

→ assume private key (X_A) of user A — $X_A < q$

cal public key (Y_A) of user A — $Y_A = \alpha^{X_A} \bmod q$

→ assume X_B — $X_B < q$

$Y_B = \alpha^{X_B} \bmod q$

$\{X_A, Y_A\}$ → Priv key KR_A, Pub key KU_A

$\{X_B, Y_B\}$ → Priv key KR_B, Pub key KU_B

Key generation

- A: $K = Y_B^{X_A} \bmod q$
- B: $K = Y_A^{X_B} \bmod q$

ex: 1) $q = 11$

$\alpha = ?$

$\alpha^1 \bmod 11$

$\alpha^2 \bmod 11 = \{1, 2, 3 \dots 10\} \Rightarrow$

$\vdots$

$\alpha^{10} \bmod 11$

$1 \Rightarrow 1^1 \quad 1^2 \quad 1^3 \quad 1^4 \quad 1^5 \quad 1^6 \quad 1^7 \quad 1^8 \quad 1^9 \quad 1^{10}$ mod

$\downarrow$ each: 1 1 1 1 1 1 1 1 1 1

$2 \Rightarrow 2^1 \quad 2^2 \quad 2^3 \quad 2^4 \quad 2^5 \quad 2^6 \quad 2^7 \quad 2^8 \quad 2^9 \quad 2^{10}$ mod

$\downarrow$ each: 2 4 8 5 10 9 7 3 6 1

no repetition
so 2 is primroot of 11

we can select 2 as α

$3 \Rightarrow$

$4 \Rightarrow$

b

we may get for other no also
we can select one amongst

$10 \Rightarrow$

$\alpha = 2$

Select $X_A \rightarrow 8 < q$

$Y_A = \alpha^{X_A} \bmod q$

$= 2^8 \bmod 11$

$= 3$

Select $X_B \rightarrow 4 < q$

$Y_B = \alpha^{X_B} \bmod q$

$= 2^4 \bmod 11$

$= 5$

User A: $\{X_A = 8, Y_A = 3\}$

User B: $\{X_B = 4, Y_B = 5\}$

Secret S — $\underline{A}$ (S) $\rightarrow$ $\underline{B}$ (R)

$K = Y_B^{X_A} \bmod q$ (pri)

$= 5^8 \bmod 11$

$K = Y_A^{X_B} \bmod q$

$= 3^4 \bmod 11$

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